

TypeScript Date Examples

Creating Dates

```
// Current date and time
const now: Date = new Date();

console.log(now);

// Specific date
const birthday: Date = new Date("2026-05-12");

console.log(birthday);

// Using year, month, day
// Month starts from 0 (0 = January)
const customDate: Date = new Date(2026, 4, 12);

console.log(customDate);
```

Getting Date Values

```
const today: Date = new Date();

console.log(today.getFullYear());
console.log(today.getMonth());
console.log(today.getDate());
console.log(today.getDay());
console.log(today.getHours());
console.log(today.getMinutes());
console.log(today.getSeconds());
```

Setting Date Values

```
const date: Date = new Date();

date.setFullYear(2030);
date.setMonth(11);
date.setDate(25);

console.log(date);
```

Formatting Dates

```
const today: Date = new Date();

console.log(today.toDateString());
console.log(today.toString());
console.log(today.toISOString());
console.log(today.toLocaleDateString());
console.log(today.toLocaleTimeString());
```

Comparing Dates

```
const date1: Date = new Date("2026-05-10");
const date2: Date = new Date("2026-05-12");

if (date1 < date2) {
    console.log("date1 is earlier");
}
```

Calculating Difference Between Dates

```
const start: Date = new Date("2026-05-01");
const end: Date = new Date("2026-05-12");

const diffMs: number = end.getTime() - start.getTime();

const diffDays: number = diffMs / (1000 * 60 * 60 * 24);

console.log(diffDays);
```

Adding Days to a Date

```
const today: Date = new Date();

today.setDate(today.getDate() + 7);

console.log(today);
```

Date Validation

```
const input: string = "2026-05-12";
const date: Date = new Date(input);

if (!isNaN(date.getTime())) {
  console.log("Valid date");
} else {
  console.log("Invalid date");
}
```

Function Example with Dates

```
function calculateAge(birthYear: number): number {
  const currentYear: number = new Date().getFullYear();
  return currentYear - birthYear;
}

console.log(calculateAge(1995));
```

Array of Dates

```
const holidays: Date[] = [
  new Date("2026-01-01"),
  new Date("2026-08-15"),
  new Date("2026-12-25")
];

holidays.forEach((holiday: Date) => {
  console.log(holiday.toString());
});
```

Mini Exercise

```
// Print current date in DD-MM-YYYY format

const today: Date = new Date();

const formattedDate: string =
  `${today.getDate()}-${today.getMonth() + 1}-${today.getFullYear()}`;

console.log(formattedDate);
```